LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

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EXP -06

6. Use Figma to simulate a handoff scenario in a mobile network and visualize the process.

Aim: -To simulate a handoff scenario in a mobile network and visualize the process using Figma.

Procedure:

1. Open Figma

2. Create a new file

3. Select the Frames

4. Fill in the content that is required for presentation

5. Design Visual Elements

6. Make it Interactive

7. Add Annotations and Explanations

8. Incorporate Multimedia

9. Storyboard Animation

10. Review and edit the Prototype

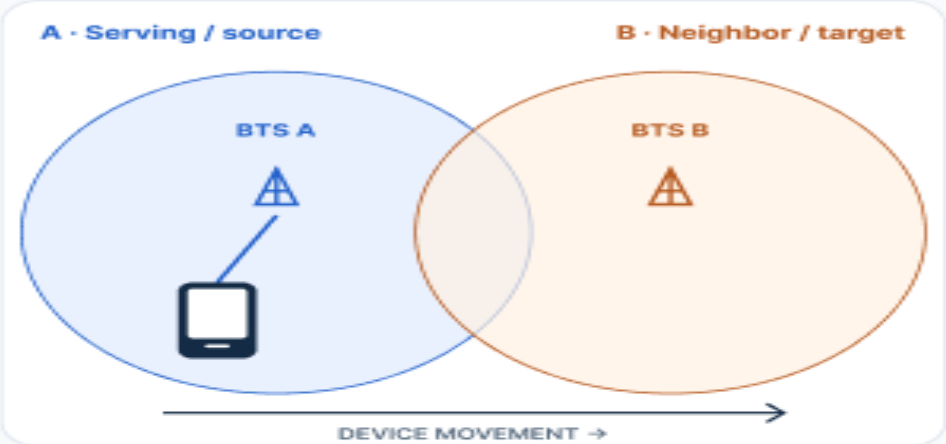
11. Save and Share

Design:

01 — THE ESSENTIAL IDEA

Handoff in a Mobile Network

An ongoing connection moves between cells to maintain service as a device moves.



Conceptual coverage, not precise cell boundaries.

One connection. A changing serving cell.

Also called handover: Cell A is the source; Cell B is the neighbor and potential target.

Hard handoff

Break-before-make.

Soft handoff

Make-before-break.

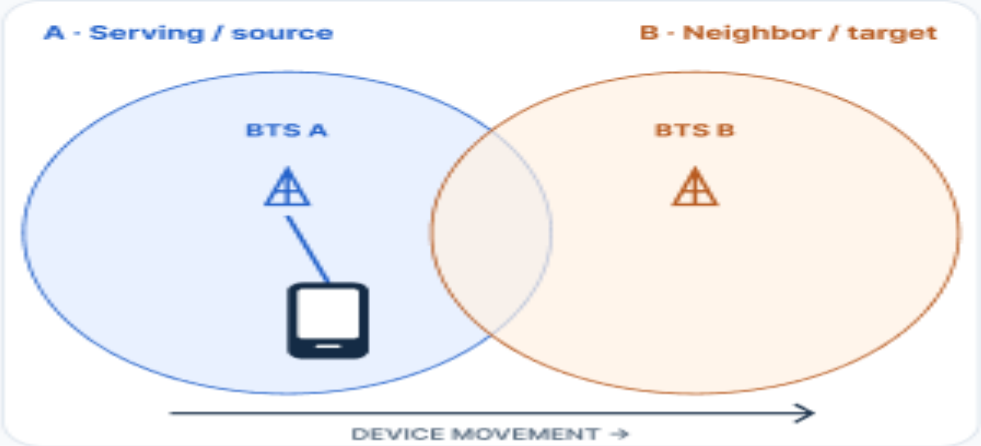
Implementation depends on the radio technology.

Illustrative educational sequence — not a live network simulation

02 — MOVE

A device on the move

The phone travels from Cell A toward the overlapping coverage of Cell B.



Conceptual coverage, not precise cell boundaries.

Early in the journey · illustrative

Cell A · 4/4
Serving

Cell B · 1/4
Target

Cell A still carries the connection

The phone can detect a neighboring cell before a handoff is needed. Movement alone does not trigger a transfer.

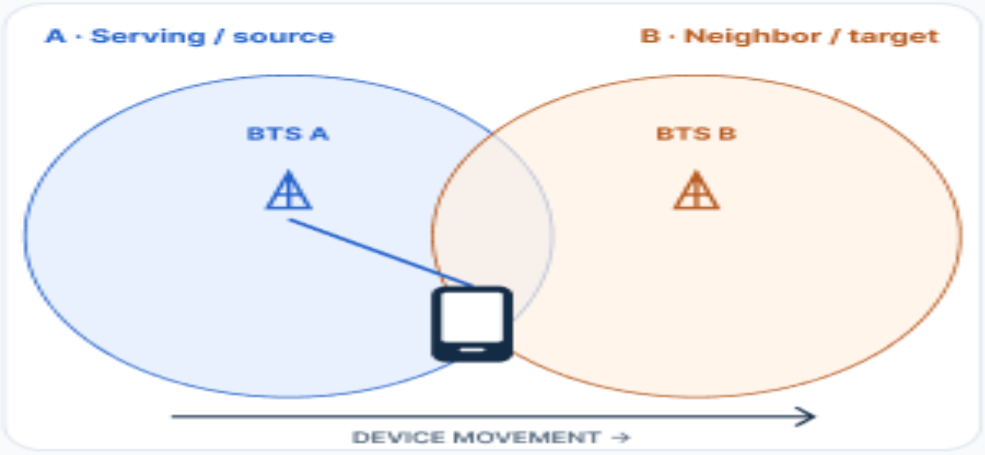
A / Serving cell
Supports the current connection.

B / Target cell
A candidate for the next connection.

03 — OBSERVE

A fades. B grows stronger.

In the overlap, the phone receives a weaker signal from A and a stronger one from B.



Conceptual coverage, not precise cell boundaries.

In the overlap · received at the phone · illustrative

Cell A · 2/4	Cell B · 4/4
Serving	Target

A: 4/4 → 2/4
Serving signal weakens

→

B: 1/4 → 4/4
Target signal strengthens

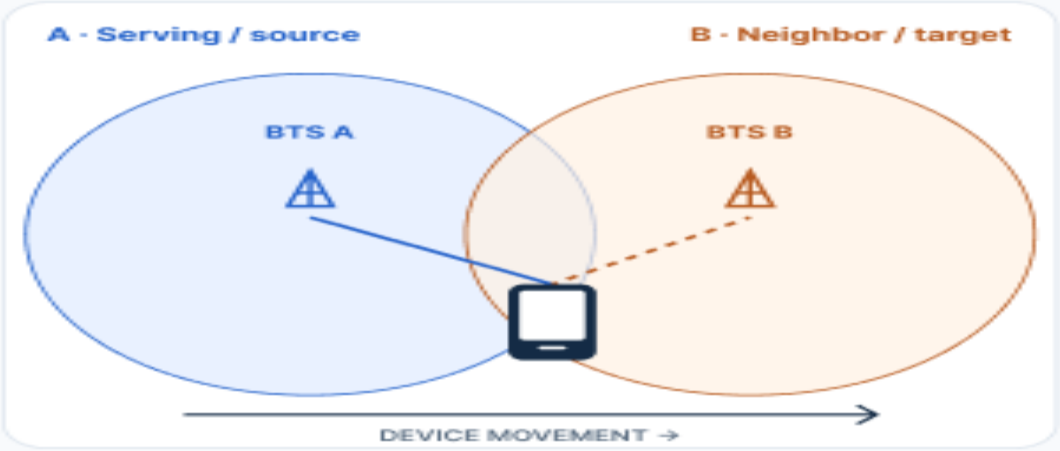
Received signal ≠ BTS transmit power

These relative bars are illustrative, not measured data. They show reception at the phone, not a change in base-station transmit power.

04 — MEASURE

Report first. Evaluate next.

The device reports measurements of its serving cell and neighboring cells to the network.



Conceptual cells; dashed = candidate, not an active link.

NETWORK EVALUATION

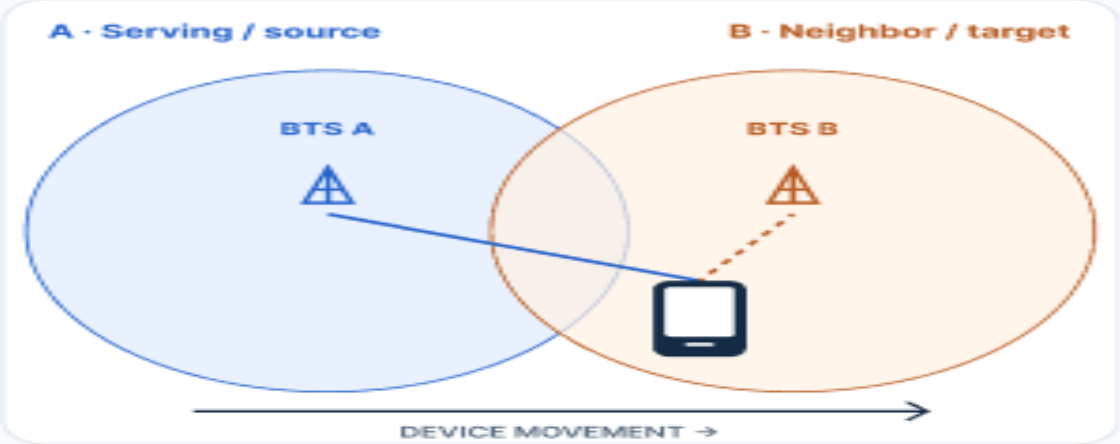
- **Signal strength & quality**
  Compare serving and neighbor measurements.
- **Thresholds & hysteresis**
  Require a sufficient margin to avoid rapid back-and-forth handovers.
- **Load & available resources**
  Network capacity may also influence the choice.

Stronger does not mean immediate

B is stronger here (A 2/4, B 4/4), but the network still evaluates the handover conditions.

05 — DECIDE & TRANSFER

Prepare the target. Transfer the link.



Conceptual cells; dashed = candidate, not an active link.

Simplified network-led sequence

Decision > **Target preparation** > **Connection transfer**

The network selects B, prepares target resources, then directs the phone to transfer its connection.

Hard handoff

Break-before-make · e.g. traditional GSM/LTE handover.

Soft handoff

Make-before-break - simultaneous radio links, e.g. CDMA/WCDMA.

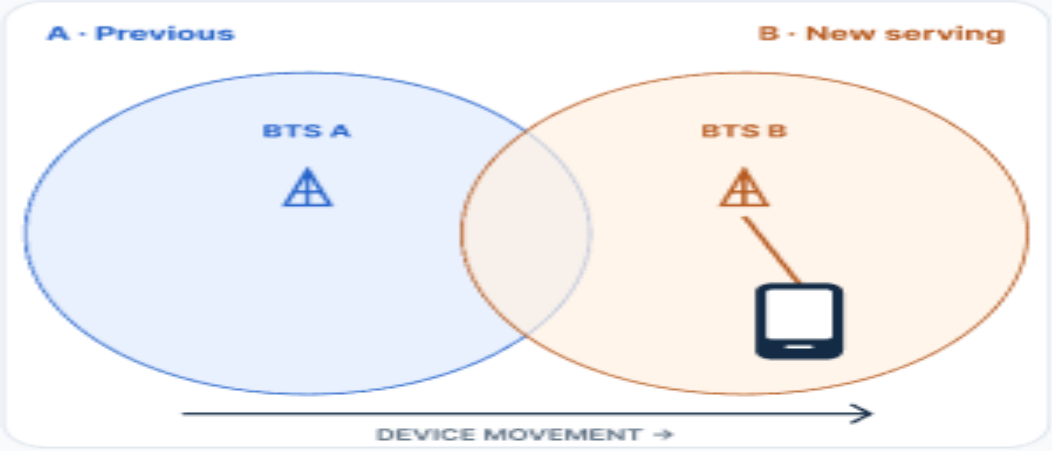
Implementation depends on the radio technology.

Soft handoff is not universal in LTE/5G. Dual connectivity is not identical to classical soft handoff.

06 — CONNECTION CONTINUES

Cell B takes over.

The handoff is complete. B is the new serving cell; A is now the previous cell.



Conceptual coverage, not precise cell boundaries.

Received at the phone - illustrative

AFTER TRANSFER	BEFORE[1]	AFTER
Cell A Previous cell	4/4	1/4
Cell B New serving cell	1/4	4/4

[1] Early journey. In the overlap: A 2/4, B 4/4.

The ongoing connection continues via B

The aim is service continuity. Any brief interruption depends on radio technology and network conditions.



Result: -
Hence the simulated a handoff scenario in a mobile network and visualized the process using Figma.